

implemented at the edge of the network in the end systems; we'll see shortly that the network-layer connection service is implemented in the routers in the network core as well as in the end systems.

Virtual-circuit and datagram networks are two fundamental classes of computer networks. They use very different information in making their forwarding decisions. Let's now take a closer look at their implementations.

4.2.1 Virtual-Circuit Networks

While the Internet is a datagram network, many alternative network architectures—including those of ATM and frame relay—are virtual-circuit networks and, therefore, use connections at the network layer. These network-layer connections are called **virtual circuits (VCs)**. Let's now consider how a VC service can be implemented in a computer network.

A VC consists of (1) a path (that is, a series of links and routers) between the source and destination hosts, (2) VC numbers, one number for each link along the path, and (3) entries in the forwarding table in each router along the path. A packet belonging to a virtual circuit will carry a VC number in its header. Because a virtual circuit may have a different VC number on each link, each intervening router must replace the VC number of each traversing packet with a new VC number. The new VC number is obtained from the forwarding table.

To illustrate the concept, consider the network shown in Figure 4.3. The numbers next to the links of R1 in Figure 4.3 are the link interface numbers. Suppose now that Host A requests that the network establish a VC between itself and Host B. Suppose also that the network chooses the path A-R1-R2-B and assigns VC numbers 12, 22, and 32 to the three links in this path for this virtual circuit. In this case, when a packet in this VC leaves Host A, the value in the VC number field in the packet header is 12; when it leaves R1, the value is 22; and when it leaves R2, the value is 32.

How does the router determine the replacement VC number for a packet traversing the router? For a VC network, each router's forwarding table includes VC

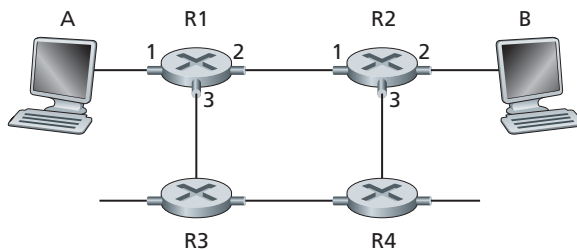


Figure 4.3 ♦ A simple virtual circuit network

number translation; for example, the forwarding table in R1 might look something like this:

Incoming Interface	Incoming VC #	Outgoing Interface	Outgoing VC #
1	12	2	22
2	63	1	18
3	7	2	17
1	97	3	87
...	...	...	...

Whenever a new VC is established across a router, an entry is added to the forwarding table. Similarly, whenever a VC terminates, the appropriate entries in each table along its path are removed.

You might be wondering why a packet doesn't just keep the same VC number on each of the links along its route. The answer is twofold. First, replacing the number from link to link reduces the length of the VC field in the packet header. Second, and more importantly, VC setup is considerably simplified by permitting a different VC number at each link along the path of the VC. Specifically, with multiple VC numbers, each link in the path can choose a VC number independently of the VC numbers chosen at other links along the path. If a common VC number were required for all links along the path, the routers would have to exchange and process a substantial number of messages to agree on a common VC number (e.g., one that is not being used by any other existing VC at these routers) to be used for a connection.

In a VC network, the network's routers must maintain **connection state information** for the ongoing connections. Specifically, each time a new connection is established across a router, a new connection entry must be added to the router's forwarding table; and each time a connection is released, an entry must be removed from the table. Note that even if there is no VC-number translation, it is still necessary to maintain connection state information that associates VC numbers with output interface numbers. The issue of whether or not a router maintains connection state information for each ongoing connection is a crucial one—one that we'll return to repeatedly in this book.

There are three identifiable phases in a virtual circuit:

- *VC setup.* During the setup phase, the sending transport layer contacts the network layer, specifies the receiver's address, and waits for the network to set up the VC. The network layer determines the path between sender and receiver, that is, the series of links and routers through which all packets of the VC will travel. The network layer also determines the VC number for each link along the path. Finally, the network layer adds an entry in the forwarding table in each router

along the path. During VC setup, the network layer may also reserve resources (for example, bandwidth) along the path of the VC.

- *Data transfer.* As shown in Figure 4.4, once the VC has been established, packets can begin to flow along the VC.
- *VC teardown.* This is initiated when the sender (or receiver) informs the network layer of its desire to terminate the VC. The network layer will then typically inform the end system on the other side of the network of the call termination and update the forwarding tables in each of the packet routers on the path to indicate that the VC no longer exists.

There is a subtle but important distinction between VC setup at the network layer and connection setup at the transport layer (for example, the TCP three-way handshake we studied in Chapter 3). Connection setup at the transport layer involves only the two end systems. During transport-layer connection setup, the two end systems alone determine the parameters (for example, initial sequence number and flow-control window size) of their transport-layer connection. Although the two end systems are aware of the transport-layer connection, the routers within the network are completely oblivious to it. On the other hand, with a VC network layer, *routers along the path between the two end systems are involved in VC setup, and each router is fully aware of all the VCs passing through it.*

The messages that the end systems send into the network to initiate or terminate a VC, and the messages passed between the routers to set up the VC (that is, to modify connection state in router tables) are known as **signaling messages**, and the protocols

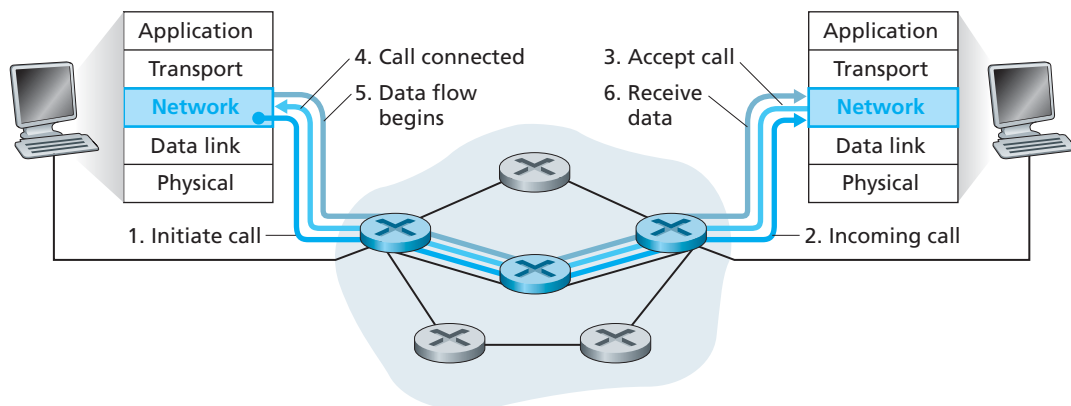


Figure 4.4 ♦ Virtual-circuit setup

used to exchange these messages are often referred to as **signaling protocols**. VC setup is shown pictorially in Figure 4.4. We'll not cover VC signaling protocols in this book; see [Black 1997] for a general discussion of signaling in connection-oriented networks and [ITU-T Q.2931 1995] for the specification of ATM's Q.2931 signaling protocol.

4.2.2 Datagram Networks

In a **datagram network**, each time an end system wants to send a packet, it stamps the packet with the address of the destination end system and then pops the packet into the network. As shown in Figure 4.5, there is no VC setup and routers do not maintain any VC state information (because there are no VCs!).

As a packet is transmitted from source to destination, it passes through a series of routers. Each of these routers uses the packet's destination address to forward the packet. Specifically, each router has a forwarding table that maps destination addresses to link interfaces; when a packet arrives at the router, the router uses the packet's destination address to look up the appropriate output link interface in the forwarding table. The router then intentionally forwards the packet to that output link interface.

To get some further insight into the lookup operation, let's look at a specific example. Suppose that all destination addresses are 32 bits (which just happens to be the length of the destination address in an IP datagram). A brute-force implementation of the forwarding table would have one entry for every possible destination address. Since there are more than 4 billion possible addresses, this option is totally out of the question.

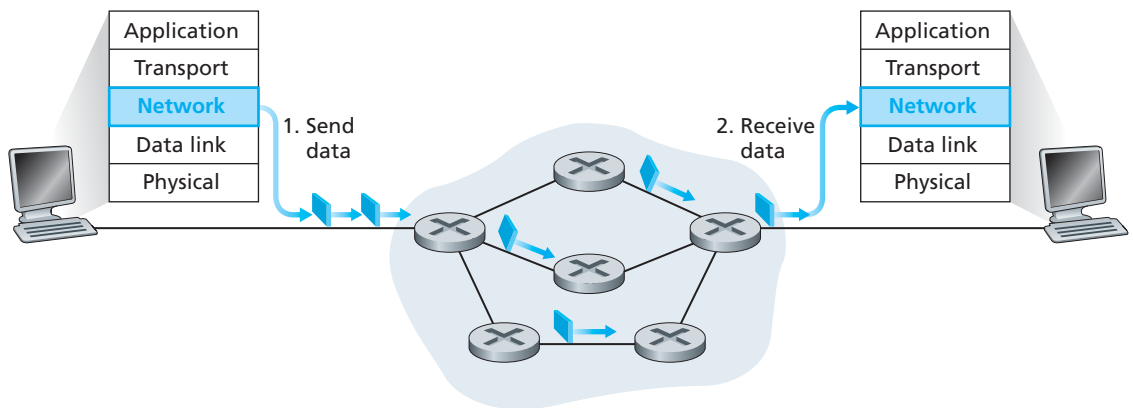


Figure 4.5 ♦ Datagram network